Point Defects in Crystals from Freysoldt *et al.*

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1 The Formation Energy as a Grand Potential

Reservoirs and the grand potential. A crystal supercell exchanges atoms of each species i with a reservoir at the chemical potential μ_i, and exchanges electrons with a reservoir at the electron chemical potential, the Fermi level E_F. The reservoirs represent the surroundings of the supercell. They fix the temperature T and the chemical potentials, and the supercell has a fixed volume V. The Legendre transform of the internal energy U with the natural variables $(T, V, \{\mu_i\}, E_F)$ is the grand potential Φ. Here, N_i denotes the number of atoms of species i and N_e the number of electrons in the pristine supercell. The grand potential of the pristine supercell is

$$\Phi_{\text{bulk}} = U_{\text{bulk}} - TS_{\text{bulk}} - \sum_i \mu_i N_i - E_F N_e = F_{\text{bulk}} - \sum_i \mu_i N_i - E_F N_e.$$

Neutral defects. Here, n_i denotes the number of atoms of species i that enter the supercell to form the defect X. A positive n_i adds atoms, for example an interstitial, and a negative n_i removes atoms, for example a vacancy. The grand potential of the supercell with the defect is

$$\Phi[X] = F[X] - \sum_i \mu_i \left(N_i + n_i\right) - E_F N_e.$$

The formation energy of the defect is the change of the grand potential at fixed $(T, V, \{\mu_i\}, E_F)$,

$$\boxed{E^f[X] \equiv \Phi[X] - \Phi_{\text{bulk}} = F[X] - F_{\text{bulk}} - \sum_i \mu_i n_i.} \tag{1.1}$$

The natural variables are the same before and after the defect forms, but $\Phi[X]$ and Φ_{bulk} describe two different systems, so the two values differ.

Charged defects. A defect in the charge state q, written X^q, has released q electrons to the electron reservoir, so the supercell holds $N_e - q$ electrons,

$$\Phi[X^q] = F[X^q] - \sum_i \mu_i \left(N_i + n_i\right) - E_F \left(N_e - q\right).$$

The terms in N_i and N_e cancel against Φ_{bulk}, and the formation energy is

$$\boxed{E^f[X^q] = F[X^q] - F_{\text{bulk}} - \sum_i \mu_i n_i + qE_F.} \tag{1.2}$$

Formation energy from DFT at 0 K. At $T = 0$, the free energy equals the internal energy, $F = U$. A DFT calculation gives U as the total energy E_{tot} of the supercell with the nuclei at rest, without the zero-point energy. The formation energy then reads

$$\boxed{E^f[X^q] = (E_{\text{tot}}[X^q] + ZPE[X^q]) - (E_{\text{tot}} + ZPE[\text{bulk}]) - \sum_i \mu_i n_i + qE_F.} \tag{1.3}$$

The two total energies come from the DFT calculations. For sufficiently heavy nuclei, the zero-point energies are neglegible. The chemical potentials μ_i and the Fermi level E_F describe the reservoirs and enter as parameters.